

Installing R and RStudio

for the Third INFOGUT Training School

Bioinformatics 101 for colon model experiments

Ubuntu Linux

The key idea: R and RStudio are two different programs. Install R first, then install RStudio. R executes your code; RStudio is the interface you will normally use during the course.

Beginner-friendly guide based on official documentation from the R Project (CRAN) and Posit.

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1. Before you start

For this course you need both R and RStudio Desktop installed on your computer before the hands-on sessions.

Software	What it does
R	The programming language and statistical computing environment. It actually runs your R code.
RStudio Desktop	A free IDE that provides a convenient interface for writing, running, organizing, and viewing R code and results.

Recommended order:

1. Install the latest stable version of R.
2. Install the latest stable version of RStudio Desktop.
3. Open RStudio.
4. Run the verification tests at the end of this guide.

Important: Do not install only RStudio. RStudio requires R to be installed on the computer first.

2. Install R on Ubuntu Linux

These instructions are for Ubuntu and follow the official CRAN Ubuntu documentation. Other Linux distributions use different installation commands.

Official CRAN Ubuntu instructions: <https://cran.r-project.org/bin/linux/ubuntu/>

Step 1 - Open Terminal

Open the Terminal application. On many Ubuntu systems you can use Ctrl + Alt + T.

Step 2 - Update package information

```
sudo apt update -qq
```

Step 3 - Install helper packages

```
sudo apt install --no-install-recommends software-properties-common dirmngr
```

Step 4 - Add the CRAN signing key

```
wget -qO- https://cloud.r-project.org/bin/linux/ubuntu/marutter_pubkey.asc | sudo tee  
-a /etc/apt/trusted.gpg.d/cran_ubuntu_key.asc
```

Step 5 - Add the CRAN repository

```
sudo add-apt-repository "deb https://cloud.r-project.org/bin/linux/ubuntu $(lsb_release  
-cs)-cran40/"
```

Step 6 - Install R

```
sudo apt install --no-install-recommends r-base
```

Step 7 - Check that R starts

R

If the R console starts, the installation worked. To exit R, type:

```
q()
```

Note: Commands beginning with `sudo` may ask for your Linux user password. When typing the password, Terminal may not display characters on screen; this is normal.

3. Install RStudio Desktop

Install RStudio only after R has been installed successfully. For the Training School, the free RStudio Desktop edition is sufficient.

Official Posit downloads: <https://posit.co/downloads/>

5. On the Posit downloads page, click the “Download RStudio” button.
6. You will be redirected to the official RStudio IDE User Guide, where a “Direct Downloads” table is displayed.
7. In the table, find the “Ubuntu 22 / 24 / 26, Debian 13” row and click the .deb package (for example, `rstudio-2026.08.2-200-amd64.deb`).
8. When the download is complete, open a Terminal in the folder containing the downloaded file and install it with:

```
sudo apt install ./rstudio-<version>-amd64.deb
```

Note: the exact RStudio version number may change. Always choose the latest stable version listed for your operating system.

Replace `<version>` with the exact filename that you downloaded. A simple method is to type “`sudo apt install ./`” and then type or paste the package filename.

If the file is in Downloads: You may first need to move to that folder with `cd ~/Downloads` and then run the installation command.

4. Open RStudio and verify the installation

Open RStudio. It should automatically detect your R installation. In the standard layout, the Console is usually in the lower-left part of the window.

Test 1 - Basic calculation

```
1 + 1
```

Expected result:

```
[1] 2
```

Test 2 - Check the R version

```
R.version.string
```

Test 3 - Show session information

```
sessionInfo()
```

Success: If these commands run without an error, R and RStudio are correctly connected and your computer is ready for the course.

5. Create and run your first R script

9. In RStudio, choose File → New File → R Script.

10. Type the following code in the script editor:

```
x <- 5
y <- 10

x + y
```

Run the current line or selected code using:

- Ctrl + Enter

You should see the result in the Console:

```
[1] 15
```

Good habit: Write and save your work in R scripts rather than typing everything directly into the Console. This makes your analysis easier to repeat and share.

6. Optional test: install an R package

R includes many functions by default. Additional functionality is distributed through packages. To test package installation, you can install ggplot2:

```
install.packages("ggplot2")
```

You normally install a package only once. To use it in a new R session, load it with:

```
library(ggplot2)
```

7. Common beginner problems

Problem	What to try
RStudio opens but cannot find R	Confirm that R was installed before RStudio, then close and reopen RStudio.
I installed R but cannot find RStudio	R and RStudio are separate programs. Download RStudio Desktop from the official Posit website.
I see several R downloads	Choose the latest stable/base release for your operating system, not a development build.
A package will not install	Read the complete error message. Some packages require additional system libraries or compilers; this is separate from the basic installation.

The .deb package is not found	Make sure Terminal is in the folder containing the downloaded file, for example <code>cd ~/Downloads</code> .
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8. Final checklist before the Training School

- ☐ R is installed.
- ☐ RStudio Desktop is installed.
- ☐ RStudio opens without an error.
- ☐ The Console shows an R version.
- ☐ `1 + 1` returns 2.
- ☐ You can create an R Script and run code from it.

Please check this before the course: Completing the installation in advance will allow us to start the hands-on R exercises directly during the Training School.

Official sources

If version numbers or system requirements change, always follow the latest information on these official pages:

CRAN - The Comprehensive R Archive Network: <https://cran.r-project.org/>

CRAN - R for Ubuntu: <https://cran.r-project.org/bin/linux/ubuntu/>

Posit - RStudio downloads: <https://posit.co/downloads/>

Posit - RStudio IDE User Guide: <https://docs.posit.co/ide/user/>